Practical: Flour, Formulas and the Protocol as Run

Part IV of the reference, the heat-driven levain bank, and a plan for maximum flavour

Sourdough Science

github.com/zrrbite/sourdough-science

What this deck covers

1. **Flour** — why protein % is not enough; what to buy (ch. 21)
2. **Formulas and method** — ratios, two pizza formulas, seven steps (ch. 22)
3. **Quick references** — home oven vs Ooni 12”; five preferments (ch. 23–24)
4. **The protocol as run** — the heat-driven levain bank, its hypotheses and open tests
5. **Holding the temperature** — the targets, and how to hold them at home
6. **A plan for maximum flavour** — seven steps aimed at acids, esters or the crust

Flags: **own result** this kitchen’s experience **hypothesis** proposed mechanism

contested the literature conflicts **not yet tested** open, not yet checked

In the kitchen: Formulas assume a domestic oven to 300°C, an Ooni 12” gas, Caputo Oro Manitoba, a pure sourdough culture kept warm, and a fridge at 3–5°C. Adjust for your own kit.

Source: README.md (equipment and conventions).

Flour

The protein number says how much, not how good

- Protein % on the bag is Kjeldahl nitrogen × a factor: the total, not the composition
- What matters: the glutenin-to-gliadin ratio and the **HMW glutenin subunit** alleles
- Glu-D1 **5+10**: longer, more cross-linked polymers — tighter mesh, more elasticity, better gas retention
- Glu-D1 **2+12**: shorter, weaker polymers — yet the same protein % on the bag
- European retail “Manitoba” is essentially unregulated, meaning roughly “high protein”: no origin or CWRS genetics guaranteed

In the kitchen: Two flours at the same protein % can make very different doughs. The label cannot tell you which.

Source: ch. 21; Payne 1987 (Glu-1 scoring, 5+10 vs 2+12); Shewry, Halford & Lafiandra 2003.

Four numbers predict whether a dough survives 48 h

Measure	What it tells you	Caputo Oro Manitoba	Generic flour
W index ($J \times 10^{-4}$)	Energy to inflate a dough sheet to bursting: strength + extensibility	W 350–380+	“Manitoba” at 13%: often W 280–310
P/L ratio	Tenacity vs extensibility; ideal ~0.5–0.7	Formulated to hit 0.5–0.7	High P: snaps back, tears. Low P: puddles on the peel
Damaged starch	Broken granules feed amylase: more sugar	Controlled by calibrated milling	Often more, from aggressive rollers
Ash	Minerals that activate amylases and proteases	~0.45–0.55% (Tipo 00 spec)	0.65–0.80%, even at the same protein

In the kitchen: The 50–80 W gap decides whether a 48 h dough reaches the oven intact. More damaged starch or ash leaves more sugar after the retard: an Ooni burn risk.

Source: ch. 21; Chopin 1927 (alveograph, W index); Barrera et al. 2013 (damaged starch).

For a long retard, buy flour by W, not protein

Choice	Spec	Why / how to use it
First choice	Caputo Oro Manitoba / premium Tipo 00: W 320+, P/L 0.5–0.7, protein 13.5–14.5%, ash <0.55%	Made for long ferments; survives 48 h with the network intact
Acceptable	Any Tipo 00 or strong bread flour with verified W 300+	No W given? Protein over 13.5% from a reputable mill: a fair, imperfect proxy
Use with caution	Generic “Manitoba” under 13% or unknown W: fine short, likely to degrade in 48 h	Retard 24 h, cut the levain (less acid, less protease), or blend in 20–30% Caputo Oro
Levain flour	Whole wheat or high-extraction	Minerals, flavour precursors, enzymes — kept to the levain’s 10–15% of the flour

In the kitchen: Flour barely matters for a 2-hour direct dough; for a 48 h retarded hybrid in an Ooni, it is the highest-impact ingredient.

Source: ch. 21.

Formulas and method

Levain up, yeast down: three starting ratios

	Conservative	Moderate	Bold
Levain (% of total flour)	10%	15%	20%
IDY (% of total flour)	0.3%	0.2%	0.15%
Levain hydration	100% (liquid)	100% (liquid)	60% (stiff)
Levain maturity	Young, just peaked	Just peaked	Peaked, slight decline OK
Acid character	Subtle background tang	Noticeable lactic roundness	Distinct, complex tang
Gluten impact	Minimal	Slight softening	Moderate — may need stronger flour
Flavour in home oven	Mild enhancement	Clear sourdough character	Pronounced, rustic

In the kitchen: More levain brings more wild yeast, so cut the IDY: at 20% with a mature culture, 0.1% may do. A stiff levain brings less water: add it to the dough.

Source: ch. 22 (home oven pizza, 250–300°C).

The Ooni formula is drier and carries far less yeast

	Home oven, 4 × ~250 g		Ooni 12”, 4 × ~275 g	
Ingredient	**Weight**	**Baker’s %**	**Weight**	**Baker’s %**
Flour, W280–320 (Ooni: Tipo 00 only)	550 g	100%	600 g	100%
Water	340 g	~62%	350 g	~58%
Liquid levain (100% hydration, at peak)	165 g	15% flour equiv.	150 g	~12.5% flour equiv.
Fine sea salt	16 g	2.9%	18 g	2.7%
Instant dry yeast	1.1 g	0.2%	0.4 g	0.06%
Olive oil (optional)	15 g	2.7%	12 g	1.8%
True flour / water, incl. the levain’s	~632 / ~422 g	**~67%**	~675 / ~425 g	**~63%**

In the kitchen: Ooni: 48 h minimum retard, non-negotiable: the yeast eats most of the sugar, so leopard spots, not burning. Under-proofed? 0.1% IDY next batch. Blown? Less, or retard longer.

Source: ch. 22. Home flour: Tipo 00 or strong bread flour.

Steps 1–4: young levain, acid in late, folds early

Step	When	Do	Why
1. Build the levain	8–12 h before mixing; room temperature	Feed 1:5:5 (culture:flour:water) with white bread flour, until just peaked: domed, bubbly, floats	Caught at or just before peak: most yeast vigour relative to LAB
2. Autolyse	30–60 min	Flour + most of the water (hold back ~30 g); mix until no dry flour; cover	Hydrates the flour and starts gluten before the levain's acid arrives
3. Final mix	—	Add levain, the rest of the water with the yeast dissolved in it, salt, oil	Mix to smooth, slightly tacky, a moderate windowpane
4. Bulk	2–3 h at ~22° C	3–4 stretch-and-fold sets, 30 min apart, in the first 90 min; then leave it	Expect +50–70% volume; the commercial yeast keeps timing predictable

Source: ch. 22 (method for both formulas).

Steps 5–7: cold retard, gentle pull, hot bake

Step	When	Do	Why
5. Ball, retard	3–5°C. Home 24–72 h; Ooni 48–72 h minimum	Home 250 g; Ooni 270–280 g (thicker, forgiving) or 220–230 g (thinner, crisp). Round tightly; oiled tubs	LAB acidify slowly, yeast nearly stops, ester precursors build. Ooni: sugar depletion
6. Temper, gentle pull	1.5–2 h, room temperature	Stretch and fold each ball 2–3 times, turning 90°; re-round loosely, no degassing	CO_2 dissolved in the cold nucleates (Henry's law): visible rise in 20–30 min
7a. Bake, home	250–300°C, 6–10 min	Preheated stone or steel	Longer bake, more roasted, malty crust aroma (ch. 14)
7b. Bake, Ooni	350–380°C stone	Preheat full flame ~15 min, then 40–50%. 270–280 g: 10–11", 2–3 min, turn every 20–30 s. 220–230 g: 11–12", 90 s–2 min	Optional final 15–20 s boost for leopard spots; 3–5 min stone recovery at full flame between pizzas

Source: ch. 22 (method); ch. 19 for the gentle pull.

Quick references

Sugar is fuel in the home oven, a burn risk in the Ooni

Parameter	Home oven (250–300°C)	Ooni 12” (350–380°C stone)
Total hydration	65–68%	63–65%
Ball weight / pie diameter	~250 g / 12”	~275 g / 10–11”
Commercial yeast (IDY)	0.2%	0.05–0.1%
Levain %	15% (moderate)	10–15%
Cold retard	24–72 h (flavour)	48–72 h (mandatory for sugar depletion)
Flame setting	N/A (thermostat)	40–50% of max (preheat at full, drop before launch)
Bake time / stone recovery	6–10 min / N/A	2–3 min + optional 15–20 s char boost / 3–5 min at full flame
Primary burn risk	Low — more browning fuel is welcome	High — residual sugar carbonises at max flame
Raw centre risk	Very low (long bake)	High at max flame — moderate flame solves this
Crust aroma (Maillard)	Extended thermal phase, rich expression	Good at 2–3 min; compressed at 60 s
Key advantage of hybrid	Flavour complexity in longer bake	Sugar depletion + flavour + cooked-through centre

Source: ch. 23.

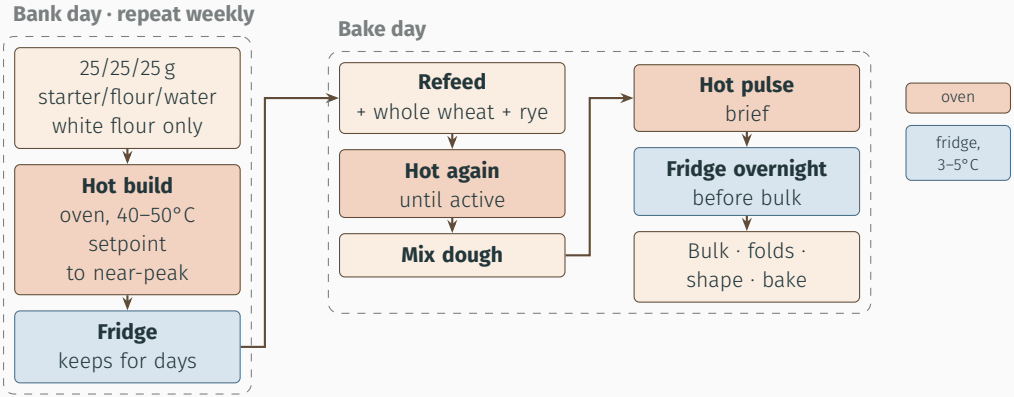
Hydration and organism set acid, flavour and gluten

	Biga	Poolish	Stiff levain	Liquid levain	Hybrid (yeast + levain)
Hydration	50–60%	100%	50–60%	100%	Any (levain separate)
Microbiology	*S. cerevisiae*	*S. cerevisiae*	Wild yeast + LAB	Wild yeast + LAB	*S. cerevisiae* + LAB
Primary acid	Minimal	Minimal	Acetic (sharp)	Lactic (mild)	Tuneable (levain type)
pH range	5.5–6.0	5.5–6.0	3.5–4.5	3.8–4.5	4.5–5.5
Flavour	Yeast aromatics	Sweet, wheaty	Sharp, vinegary	Round, creamy	Complex esters, balanced tang
Gluten effect	Elastic, tight	Extensible, soft	Extensible + protease	Extensible + protease	Mostly elastic, slight acid softening
Crumb	Fine, uniform	Open, regular	Wild, irregular	Open, moderate	Controlled open, uniform
Ferment time	12–24 h	8–16 h	6–12 h (cycle)	4–8 h (cycle)	2–3 h bulk + cold retard
Best for	Ciabatta, focaccia	Baguettes, Neapolitan pizza	Country bread	Enriched doughs	Home oven pizza
Pizza suitability	Good	Excellent (traditional)	Viable	Good	Optimal for home oven

Source: ch. 24 — *"Same vessel, same hydration, completely different ecology. Or: combine them."*

The protocol as run

Hot, cold for days, hot, hot, cold overnight, then bulk



In the kitchen: Bank several jars at once; on bake day, one jar refed with wholegrain is the levain. This is how the bread is made here now **own result**; why it works is still hypothesis.

Source: notes/heat-driven-levain-bank.md.

What changed: the cold rest now comes before bulk

	Chapter 08 protocol	This protocol
Levain	Built hot, used at once	Built hot, banked cold for days, refed hot
Wholegrain	In the build	Withheld until the bake-day refeed
Hot dough phase	2 h bulk at 40°C+	Brief pulse straight after mixing
Cold phase	After bulk (retard)	Before bulk (enzymatic rest), optionally also after
Bulk	Hot	Room temperature, after the cold rest

- The cold-before-bulk step is the one that changes the character of the result **own result**
- The idea, from the *Double Enzymatic Activation* notes: amylase works overnight while the microbes are dormant, so the dough reaches bulk pre-sweetened and extensible
- Here that rest is stacked on a hot front end instead of replacing it

Source: notes/heat-driven-levain-bank.md.

Every hot session follows a fresh feed and a rest

hypothesis a mechanism proposed in the note, not yet tested

- Ch. 11: the most useful defence against heat stress is to **feed right before a hot session** — most substrate, least acid and ethanol, a full ATP budget
- This protocol does it at every hot step without being designed to: the bank build, the bake-day refeed, and the dough mix (the largest feed of all)
- Ch. 11 frames safe heat as brief excursions with recovery between; days at 3–5° C between build and refeed are a long recovery window
- So each hot pulse meets a culture that is rested, not depleted

In the kitchen: If you heat a levain, feed it just before. A ripe, sour one carries the acid and ethanol that make heat more lethal.

Source: notes/heat-driven-levain-bank.md; ch. 11.

White flour in storage, wholegrain at activation

hypothesis a mechanism proposed in the note, not yet tested

In the fridge: keep it inert

- The enemy is over-acidification
- Acid and ethanol make the *next* heat exposure more lethal (ch. 11)
- White flour has less amylase, fewer minerals and less LAB fuel, so the jar sours more slowly

On bake day: load it

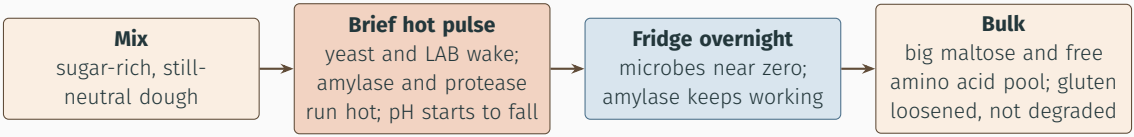
- The goal flips: most enzyme fuel, most microbial kick
- Whole wheat and rye: amylase, protease, minerals, lysine-rich protein for the crust
- Fuel arrives just before two hot pulses and the overnight rest, not burnt off in the fridge

In the kitchen: Test: bank one jar with 10% rye and one white-only; refeed and heat side by side; compare smell, rise and heat tolerance. **not yet tested**

Source: notes/heat-driven-levain-bank.md.

Hot pulse for the enzymes, fridge to still the microbes

hypothesis a mechanism proposed in the note, not yet tested



- Enzyme rate roughly doubles per 10°C (Q_{10}): a dough at pH 5.5 and 40°C has proteolytic throughput comparable to one at pH 4.5 and 25°C
- $Q_{10} \approx 2$ is a general rule of biochemistry, an order-of-magnitude guide, not a dough measurement
- A plausible reason the dough comes out feeling “truly enzymatically activated” **own result**

Source: notes/heat-driven-levain-bank.md; notes/proteolysis-ph-heat.md; ch. 25 (note on Q_{10}).

The risk is a slack dough, and four levers limit it

- An early cold rest relies on a high pH: protease stays quiet, only amylase works
- A hot pulse before the fridge erodes that: heat partly substitutes for low pH in driving protease
- The failure mode: the dough leaves the fridge slack rather than extensible

Brief pulse	Low levain	Strong flour	Salt at the mix
Ch. 07: no more than 30–60 min for the dough itself	10–15%	W 350+	Salt is the largest single inhibitor of cereal protease

In the kitchen: Here the overnight rest beats going straight to bulk **own result**, which suggests the balance is on the right side — with this flour.

Source: notes/heat-driven-levain-bank.md; ch. 07; notes/proteolysis-ph-heat.md.

The oven air is 45°C; the levain core is still unknown

- “Lots of heat” is an oven setpoint, not the dough temperature (ch. 10)
- 75 g of levain in a glass jar ramps slowly in air; its core likely sits well below the setpoint for most of the session **hypothesis**
- A slow ramp protects better than a sudden shock (ch. 11): that could reconcile the protocol with ch. 07’s 40–45°C viability-decline range
- **Measured 2026-09-13:** empty-oven air ~45°C on the usual setting **own result**. A spot reading, not a logged hold, and not a loaded oven
- 45°C is where ch. 07 moves from “viability decline” to “thermal kill territory for most yeast”, so the explanation must come from the core

In the kitchen: The honest statement: the air is measured at 45°C, the core temperature is unknown, and the bread rises.

Source: notes/heat-driven-levain-bank.md; ch. 07, 10, 11; Guyot et al. (slow ramps); Sadot et al. 2017 (core lag).

Holding the temperature

Measure the dough, not the air

What	Temperature
New starter, first two weeks	24–27°C
Reviving a dried starter	24–26°C
Tripling test (“done”)	27°C
Hot levain build	35–38°C
Levain bank, oven setpoint	40–50°C
Warm autolyse, 60–90 min	38–42°C
Bulk, hybrid formulas	≈ 22°C
Cold retard	3–5°C

- Air is a poor conductor: a jar or tub lags the air, more for bigger masses
- Here the oven air is measured at ≈ 45°C; the levain core never has been
- A dual-probe thermometer, air and core: the gap is the number that matters
- The ramp matters too: a hot water bath is harsher than it looks

In the kitchen: Whatever heats it, put a probe in the dough.

Source: notes/holding-temperature.md; ch. 07, 10, 11, 14, 17, 22; starter note; levain bank note

Free and improvised: fine for a starter, blind for hot builds

Method	Range	Best for, and the catch
Oven, light on	what the lamp gives; 27°C in the video	a starter at 27°C; LED lamps barely heat
Oven, low setting	≈ 45°C air here	hot builds; dials are unreliable low, so probe the core
Residual heat	falling	short boosts; no control, overshoots easily
Water bath	up to hot tap; exact with a sous-vide stick	a steady starter; fast ramp, so not for hot builds
Warm spots	a few °C above the room	a cold day: a mug of boiling water in a closed microwave
Fridge	3–5°C, measure yours	retards

In the kitchen: Leave a note on the oven whenever something is proving in it.

Source: notes/holding-temperature.md; starter note; levain bank note; ch. 11

Boxes that hold a setpoint

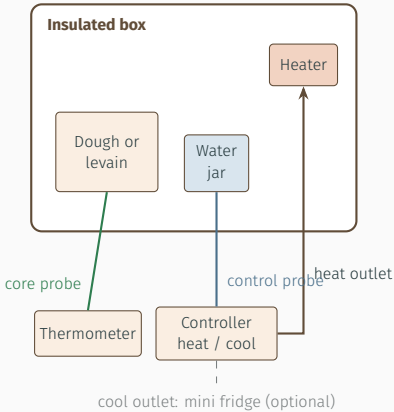
Box	Range	Best for, and the catch
Controller in a box	what the heater reaches	any target; see the next slide
Chamber Stage 0	to ≈ 45°C	any run you want measured: the only one that logs air and core
B&T Folding Proofer	21–35°C humid, 21–49°C dry	doughs, covered warm autolyse, levain to 49°C; heats from below, does not cool
B&T Sourdough Home	5–50°C, heats and cools	one starter jar up to 1 L, held within 0.5–1.5°C

In the kitchen: Starter: the oven light if it really gives 24–27°C, else a water bath or a box. Hot builds: the oven with a core probe, or Stage 0 for a record.

Source: notes/holding-temperature.md; Brød & Taylor FAQ (manufacturer's figures); chamber/stage0

A controller in a box holds any target

- **Controller:** e.g. Inkbird ITC-308: heat and cool outlets, alarms
- **Heater:** 40–60 W ceramic emitter in a ceramic socket, or a seedling mat; an LED bulb makes no heat
- **Box:** cooler, styrofoam or a dead fridge; ≈ 16 W holds 45° C in 35 L
- **Probe:** in the air or a water jar the dough's size; not in the dough **hypothesis**
- **Safety:** guard, air gap, aim along a wall, never at the dough



In the kitchen: The controller holds the air; a separate core probe shows what the dough did.

Source: notes/holding-temperature.md; Inkbird ITC-308 specifications (manufacturer); chamber/stage0

A plan for maximum flavour

Flavour has three sources; each step works on one

Acids from the bacteria	**Esters** from the yeast	**Crust browning** in the oven
1 Whole-grain levain, stiffer for sharpness	**4** Keep the yeast active; handle gently late on	**3** Warm autolyse, no salt
2 More levain in bread: 20–30%	**+** A pinch of commercial yeast	**5** Cold retard the shaped loaf
		6 Bake darker than feels right
		7 Eat the crust the same day

In the kitchen: Change one thing per bake and taste it against your usual loaf. Several of these steps are bets, not measurements.

Source: the flavour plan, synthesised from ch. 04, 07, 12–14, 16, 17.

Steps 1–2: a stiffer, bigger levain for more acid

1. Levain: whole grain, and stiffer if you want sharpness

- Rye or wholemeal in the bake-day feed, as the levain bank already does
- A stiff levain (50–60% water) shifts the acid toward acetic, which is sharper (ch. 04)
- Not to extremes: enough acetic acid stops sourdough yeast growing altogether — complete inhibition at 150 mmol/L undissociated acetic acid — and the yeast makes the esters

2. More levain in bread: 20–30%

- More acid, and more of the bacteria's protein-cutting enzymes; commits to a sourer loaf (ch. 17)
- Tension: the levain bank holds levain to 10–15% so a hot-pulsed dough won't go slack

In the kitchen: Raise levain or stiffness one bake at a time. Pizza ratios stop at 20% (ch. 22); 20–30% is for bread.

Source: ch. 04, 17; Gänzle et al. 1998 (acetic acid and sourdough yeast); notes/heat-driven-levain-bank.md.

Steps 3–4: free amino acids early, keep the esters in

3. Warm autolyse, no salt: 60–90 min at 38–42°C, then add levain and salt

- Frees amino acids, which become the crust's malty, roasted aroma (ch. 17) **not yet tested**
- The levain bank's hot pulse and overnight cold rest do the same job **hypothesis**

4. Keep the yeast active, and handle gently late on

- Esters are made while the yeast works (ch. 13)
- A warm bulk may raise them, but the evidence conflicts: check by taste (ch. 07) **contested**
- Fold early, handle gently late: escaping CO_2 probably carries esters away (ch. 16) **hypothesis**
- A pinch of commercial yeast adds a large ester-making population without losing the rise (ch. 12–14)

Source: ch. 07, 12–14, 16, 17; Calvel et al. 2001 (autolyse); Kucharczyk & Tuszyński 2018 vs Ferrández-Gómez et al. 2024 (esters and heat).

Steps 5–7: retard cold, bake dark, eat it the same day

5. Cold retard the shaped loaf: 12–36 h at 3–5°C

- Flour enzymes keep making sugar and amino acids for the crust, and the acid keeps building (ch. 14)
- Stop before the sugar runs out: sugar is what browns

6. Bake darker than feels right

- Most of a loaf's aroma forms in the bake (pumpernickel, Majcher et al. 2020); acid slows browning
- Steam for about 20 min at 250°C, then bake uncovered until deep brown **not yet tested**

7. Eat the crust the same day

- The roasty crust aroma is believed to fade within hours **not yet tested**; the sourness holds

Source: ch. 13, 14; Majcher et al. 2020. The bake times and temperatures are general practice, not from the repo.

Takeaways

- Buy flour by W and P/L, not protein %: for a 48 h retard, W 320+ and P/L 0.5–0.7
- Levain up, yeast down: pizza takes 10–20% levain, bread tolerates 20–30%
- In the Ooni, a 48 h minimum retard uses up the sugar that would burn; a 40–50% flame cooks the centre
- Hydration and organism pick the preferment: stiff is acetic, liquid is lactic, the hybrid is tuneable
- The levain bank — hot builds after every feed, a cold rest before bulk — works here **own result**; why it works is still hypothesis, and the core temperature is unmeasured
- Flavour is acids, esters and crust: change one thing per bake and taste

Open questions

The levain bank not yet tested

1. Log core temperature in a banked jar and in the dough during the hot pulse
2. Bank life: how many fridge days before the refeed lags?
3. Does the hot build double? Doubling shows gas, not cell division; serial hot passage decides
4. 10% rye vs white-only at storage
5. With and without the overnight rest
6. Hot pulse of 30, 60, 90 min: where does extensible turn slack?

Flavour and wider threads

- Warm bulk and esters: taste side by side against a room-temperature control (ch. 07) contested
- The Strecker/pyrazine crust claim: the least directly evidenced in the repo; wants a sensory comparison (ch. 17)
- The plan's bake times, and how fast crust aroma fades not yet tested
- Every temperature here is still a setpoint, not a reading: the logging chamber (`chamber/`) is designed, first stage built, not yet instrumented

Source: notes/heat-driven-levain-bank.md (open tests); README.md (status and open threads); the flavour plan.

Sources

Flour

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- **Barrera GN et al. (2013)** Effect of damaged starch on the rheological properties of wheat starch suspensions. *J Food Eng* 116(1):233–239.

Heat and the levain

- **Guyot S, Ferret E, Gervais P** Thermal death rate of *Saccharomyces cerevisiae*: dependence on the kinetics of heat perturbation.
- **Sadot M et al. (2017)** Numerical modelling of convective heat transfer during bread baking. *J Food Eng*.

Sources (cont.)

Flavour

- **Calvel R, Wirtz RL, MacGuire JJ (2001)** *The Taste of Bread*. Aspen.
- **Majcher MA et al. (2020)** Formation of key aroma compounds during preparation of pumpernickel bread. *J Agric Food Chem* 68(38):10352–10360.
- **Kucharczyk K, Tuszyński T (2018)** The effect of temperature on fermentation and beer volatiles at an industrial scale. *J Inst Brew* 124(3):230–235.
- **Ferrández-Gómez B et al. (2024)** Sourdough yeast strains exhibit thermal tolerance, high fermentative performance, and a distinctive aromatic profile in beer wort. *Foods* 13.
- **Gänzle MG, Ehmann M, Hammes WP (1998)** Modeling of growth of *L. sanfranciscensis* and *C. milleri* in response to process parameters of sourdough fermentation. *Appl Environ Microbiol* 64(7):2616–2623. The 150 mmol/L acetic acid limit; full entry in the starter note.

In the repo

- docs/21–25; notes/heat-driven-levain-bank.md; notes/proteolysis-ph-heat.md; the flavour plan (from ch. 04, 07, 12–14, 16, 17).